DREAM-MoE: Downstream Routing Error-Aware Margin-Preserving Quantization for Mixture-of-Experts Large Language Models

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Abstract

Post-training quantization (PTQ) is essential for deploying large Mixture-of-Experts (MoE) language models, but quantization can perturb router scores and change the selected experts. Existing router-aware quantization methods mainly align the router outputs of the quantized model with those of the full-precision model at the same layer. However, even small errors in matching router logits can change the relative ordering of experts and alter the selected top-k set. Moreover, in block-wise PTQ where each transformer decoder block is quantized sequentially, reconstruction errors from the currently quantized decoder block can perturb the input to the following MoE router, leading to downstream routing changes. We propose DREAM-MoE, a downstream routing error-aware margin-preserving objective for MoE quantization. DREAM-MoE preserves expert orderings that directly affect top-k routing decisions by enforcing pairwise margins among selected experts and between selected experts and near-boundary unselected experts. It further regularizes each quantized decoder block using the downstream routing behavior of the next MoE router. DREAM-MoE is an auxiliary calibration objective and can be integrated into learning-based PTQ frameworks without adding inference-time modules or computational overhead. Across three MoE LLMs under 4-bit and 3-bit weight-only quantization, DREAM-MoE achieves the best average downstream accuracy in five of six model-bit settings and the lowest language-modeling perplexity in five of six settings.

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1. Introduction

Post-training quantization (PTQ) reduces the memory footprint and inference cost of LLMs by representing weights, and sometimes activations, with low-bit formats. This is particularly important for Mixture-of-Experts (MoE) LLMs, whose total parameter count can be large even though only a small subset of experts is activated per token (Shazeer et al., 2017; Fedus et al., 2022). However, MoE quantization is uniquely sensitive to discrete routing: small perturbations to router scores can change the selected top-k experts, causing the quantized model to execute a different expert set from the full-precision model. We refer to this quantization-induced change in expert selection as *expert-shift*. Existing MoE quantization methods mitigate expert-shift by aligning router logits, probabilities, top-k scores, ranks, or score gaps (Fu et al., 2025; Chen et al., 2025; Fang & Huang, 2025). Yet such logit-value matching does not directly guarantee expert ordering, and in block-wise PTQ over transformer decoder blocks, reconstruction errors from the currently quantized block can perturb the following MoE router's decision.

Motivated by these observations, we propose *DREAM-MoE*, a **D**ownstream **R**outing **E**rror-**A**ware **M**argin-Preserving objective for **MoE** quantization. DREAM-MoE is not a standalone quantizer, but an auxiliary calibration objective that can be integrated into learning-based PTQ frameworks such as AutoRound (Cheng et al., 2024), OmniQuant (Shao et al., 2024), and EAC-MoE (Chen et al., 2025). It replaces logit-value matching with a pairwise margin-preserving loss over routing-critical expert pairs, and extends the same principle to downstream routing so that each quantized block better preserves the next MoE block's expert selection.

Empirically, DREAM-MoE consistently improves quantized MoE models under the same quantization format and optimization budget. Across three MoE LLMs and two weight-only quantization settings, DREAM-MoE achieves the best average downstream accuracy in five of six model-bit configurations and the lowest language-modeling perplexity in five of six settings. At 4-bit, DREAM-MoE improves average downstream accuracy over the strongest baseline by $+1.30$ points on average. At 3-bit, where quantization noise is larger, DREAM-MoE obtains the best perplexity for all three models and improves average downstream accuracy on two of the three models. These results suggest that preserving routing-critical margins, especially for downstream routers affected by block-wise quantization, is an effective way to improve MoE PTQ.

Our contributions are threefold: (i) we show that router-logit matching does not necessarily preserve expert ordering or top-k membership; (ii) we introduce a margin-preserving pairwise routing objective focused on selected and near-boundary experts; and (iii) we extend this objective to downstream routing, so that each quantized block is optimized not only for local reconstruction but also for preserving the next MoE block's expert selection.

2. Related Work

An MoE layer uses a router to select a predefined number k of experts for each token. Given a token representation x, the router computes expert scores $r(x)$ and selects the k experts with the largest scores:

$$T(x) = \mathrm{TopK}(r(x), k). \tag{1}$$

The MoE layer output is then

$$y(x) = \sum_{e \in T(x)} g_e(x) E_e(x), \tag{2}$$

where E_e denotes expert e, and $g_e(x)$ is the corresponding routing weight. This sparse computation makes MoE quantization particularly sensitive to routing errors. If quantization changes $T(x)$, the model does not merely suffer a small activation perturbation; it executes a different subset of experts.

Motivated by this routing sensitivity, recent MoE quantization methods have introduced router-aware objectives for learning quantization parameters or calibrating routers. EAQuant aligns router logits and routing distributions between the original and quantized models (Fu et al., 2025). EAC-MoE improves expert-selection consistency by aligning top-ranked router outputs during quantization (Chen et al., 2025). ExpertQuant proposes two router-alignment losses: Rank-Aware Jaccard (RAJ) Loss, which aligns ordered top-k expert lists, and Gap Hinge (GH) Loss, which preserves consecutive score gaps within the original top-k experts (Fang & Huang, 2025). These works demonstrate that routing preservation is essential for MoE quantization.

DREAM-MoE differs from these methods in two aspects. First, rather than aligning only the router output at the current block, our method considers how the quantized output of the current decoder block affects the routing decision in the following MoE block. Second, while GH preserves gaps only within the original top-k experts, DREAM-MoE also considers expert pairs around the top-k cutoff, where low-ranked selected experts compete with nearby outsider experts. This directly targets the top-k decision boundary, where quantization errors can cause previously unselected experts to enter the selected set.

3. Downstream Routing Error-Aware Margin-Preserving MoE Quantization

We introduce DREAM-MoE, an auxiliary objective for MoE quantization. Unless otherwise specified, we use "block" to refer to the unit being quantized, which may be a decoder block or a layer depending on the quantization method. DREAM-MoE is designed to be integrated into calibration-based quantization methods that optimize quantization parameters, such as scales, zero-points, or rounding offsets, using calibration data. Given the standard reconstruction objective, which minimizes the output discrepancy between a quantized block and the corresponding original block, DREAM-MoE adds routing-aware losses that encourage the quantized model to preserve the expert-selection behavior of the original model.

The proposed objective consists of two components. First, we define the core routing alignment objective used in DREAM-MoE, which preserves the original model's expert ordering by matching pairwise margins within the top-k experts and around the top-k selection boundary, rather than directly matching router logits with MSE (Section 3.2). Second, we apply the same ordering-preservation principle to downstream routing by considering how the quantized output of the current block affects the routing result of the next block (Section 3.3). The final training objective combines the standard reconstruction loss with these routing-aware terms.

3.1. Overall Objective

We first define the standard block reconstruction objective. Let $h_{\ell,t}^{\mathrm{orig}}$ and $h_{\ell,t}^{q}$ denote the output of block ℓ for token t in the original and quantized models, respectively. The token-level reconstruction loss is

$$\mathcal{L}_{\mathrm{rec}}^{(t)} = \left\| h_{\ell,t}^{q} - h_{\ell,t}^{\mathrm{orig}} \right\|_2^2. \tag{3}$$

For a calibration mini-batch $\mathcal{B}$, we use

$$\mathcal{L}_{\mathrm{rec}} = \frac{1}{|\mathcal{B}|} \sum_{t \in \mathcal{B}} \mathcal{L}_{\mathrm{rec}}^{(t)}. \tag{4}$$

DREAM-MoE augments this reconstruction objective with two routing terms: a current-router alignment loss and a downstream-router alignment loss. The downstream term is used adaptively, since quantizing a given block does not always substantially change the routing behavior of the following block. We therefore compute a downstream routing overlap $\sigma_{\ell\to\ell+1}$, which measures how much the top-k expert set of the router in block $\ell+1$ is preserved after quantizing block ℓ. The downstream loss is activated only when this overlap is below a threshold τ.

The per-block objective is

$$\mathcal{L}_\ell = \mathcal{L}_{\text{rec}} + \lambda_1 \mathcal{L}_{\text{router}} + \mathbb{I}[\sigma_{\ell\to\ell+1} < \tau]\lambda_2 \mathcal{L}_{\text{dr}}. \tag{5}$$

Here, $\mathbb{I}[\cdot]$ is an indicator function that returns 1 when the condition is true and 0 otherwise. $\mathcal{L}_{\text{router}}$ is applied only when block ℓ has a router, and $\mathcal{L}_{\text{dr}}$ is applied only when block $\ell+1$ has a downstream MoE router and its routing overlap satisfies $\sigma_{\ell\to\ell+1} < \tau$. For the last decoder block, where no downstream router exists, $\mathcal{L}_{\text{dr}}$ is set to zero.

3.2. Router Alignment Loss

Limitation of TopM-MSE. When learning quantization parameters, a common way to encourage router alignment is to add an auxiliary TopM-MSE loss on router logits. For each token, let $\mathbf{r}_{\text{orig}} \in \mathbb{R}^E$ and $\mathbf{r}_q \in \mathbb{R}^E$ denote the router logit vectors of the original and quantized models, respectively, where E is the number of experts. We write $r_{\text{orig}}(e)$ and $r_q(e)$ for the scalar logit assigned to expert e. TopM-MSE first selects the m experts with the largest router logits under the original router and then matches the original and quantized logits at those same expert indices:

$$\mathcal{L}_{\text{TopM}} = \frac{1}{m}\sum_{e\in\mathcal{M}} \left(r_q(e) - r_{\text{orig}}(e)\right)^2, \tag{6}$$

where $\mathcal{M}$ is the set of m expert indices with the largest values of $r_{\text{orig}}(e)$, k is the number of routed experts per token, and $m \geq k$. In practice, m is often chosen to be larger than k so that the loss aligns not only the selected top-k experts but also near-boundary experts that may enter the selected set after quantization (Chen et al., 2025).

This auxiliary loss, however, matches logit values rather than directly preserving expert ordering. For example, consider a two-expert router that selects a single expert, i.e., $k = 1$. Let $\mathbf{r}_{\text{orig}} = [1.0, 0.0]$, so the original router selects expert 1. The two quantized candidates $\mathbf{r}_q^{(A)} = [2.0, 1.0]$ and $\mathbf{r}_q^{(B)} = [0.0, 1.0]$ have the same squared error:

$$\|\mathbf{r}_q^{(A)} - \mathbf{r}_{\text{orig}}\|_2^2 = \|\mathbf{r}_q^{(B)} - \mathbf{r}_{\text{orig}}\|_2^2 = 2. \tag{7}$$

However, $\mathbf{r}_q^{(A)}$ preserves the original ordering and therefore selects expert 1, whereas $\mathbf{r}_q^{(B)}$ flips the ordering and selects expert 2. Thus, the same TopM-MSE value can correspond to different routing decisions.

Pair construction. For each token t, we form expert pairs from the original-rank sorted expert indices. Let $\pi_t(a)$ denote the expert with rank a under the original router, where rank 0 is the highest-scoring expert. DREAM-MoE uses two groups of pairs.

Adjacent top-k pairs. We include the $k-1$ consecutive pairs inside the original top-k experts:

$$\mathcal{P}_{\text{adj}}(t) = \{(\pi_t(a), \pi_t(a+1)) : 0 \leq a < k-1\}. \tag{8}$$

These pairs preserve the local ordering among experts selected by the original model.

Boundary-pool pairs. We also form a boundary pool around the top-k cutoff. The pool contains the b_{sel} lowest-ranked selected experts and the b_{out} closest outsider experts:

$$\mathcal{I}_t = \{\pi_t(a) : k - b_{\text{sel}} \leq a \leq k + b_{\text{out}} - 1\}. \tag{9}$$

We use all ordering pairs within this pool:

$$\mathcal{P}_{\text{bnd}}(t) = \{(\pi_t(a), \pi_t(b)) : a < b,\ \pi_t(a), \pi_t(b) \in \mathcal{I}_t\}. \tag{10}$$

These pairs target the selection boundary, where quantization is most likely to change top-k membership. The final pair set is

$$\mathcal{P}(t) = \mathcal{P}_{\text{adj}}(t) \cup \mathcal{P}_{\text{bnd}}(t). \tag{11}$$

For each pair $(i, j) \in \mathcal{P}(t)$, the original ranking satisfies $r_{\text{orig}}^{(t)}(i) \geq r_{\text{orig}}^{(t)}(j)$. We define the original margin as

$$m_{\text{orig}}^{(t)}(i,j) = r_{\text{orig}}^{(t)}(i) - r_{\text{orig}}^{(t)}(j), \tag{12}$$

and the quantized margin as

$$m_q^{(t)}(i,j) = r_q^{(t)}(i) - r_q^{(t)}(j). \tag{13}$$

We then define the margin violation as

$$z_{ij}^{(t)} = \eta m_{\text{orig}}^{(t)}(i,j) - m_q^{(t)}(i,j), \tag{14}$$

where $\eta \in (0, 1]$ is a margin relaxation factor. The token-level router alignment loss is

$$\mathcal{L}_{\text{router}}^{(t)} = \frac{1}{|\mathcal{P}(t)|}\sum_{(i,j)\in\mathcal{P}(t)} \text{softplus}(z_{ij}^{(t)}). \tag{15}$$

This loss acts as a smooth margin-preserving ranking penalty. If the quantized router preserves the original ordering and maintains a margin close to that of the original model, then $m_q^{(t)}(i,j) \geq \eta m_{\text{orig}}^{(t)}(i,j)$ and $z_{ij}^{(t)}$ becomes non-positive, resulting in a small softplus penalty. In contrast, if quantization shrinks the margin too much or reverses the order of the pair, $z_{ij}^{(t)}$ becomes positive and the loss increases. Thus, the objective does not require exact logit matching; instead, it encourages the quantized router to preserve the relative ordering and a relaxed version of the original margin for routing-critical expert pairs.

The mini-batch current-router loss is

$$\mathcal{L}_{\text{router}} = \frac{1}{|\mathcal{B}|}\sum_{t\in\mathcal{B}} \mathcal{L}_{\text{router}}^{(t)}. \tag{16}$$

3.3. Downstream Router Alignment

Why downstream routing matters. The block reconstruction loss alone may admit many quantization parameter choices with similar reconstruction error. These choices can produce different output representations and therefore different routing decisions in the next MoE block. Thus, among reconstruction-equivalent solutions, DREAM-MoE prefers the one whose output is less disruptive to the downstream router.

Let $A^{\mathrm{orig}}_{\ell+1}$ denote the prefix of block $\ell+1$ up to the router input. This prefix includes all operations in the next block before the MoE router, such as normalization, self-attention, residual addition, and the pre-MoE normalization. Let $R^{\mathrm{orig}}_{\ell+1}$ be the router of block $\ell+1$ in the original model. For the original and quantized outputs of block ℓ, we compute downstream router logits as

$$\mathbf{r}^{\mathrm{orig}}_{\ell+1} = R^{\mathrm{orig}}_{\ell+1}\Big(A^{\mathrm{orig}}_{\ell+1}(h^{\mathrm{orig}}_{\ell})\Big), \tag{17}$$

and

$$\mathbf{r}^{q}_{\ell+1} = R^{\mathrm{orig}}_{\ell+1}\Big(A^{\mathrm{orig}}_{\ell+1}(h^{q}_{\ell})\Big). \tag{18}$$

The next-block prefix and router serve as a fixed routing reference; their parameters are not updated.

Downstream router loss. We define the downstream alignment loss by applying the same pair construction and pairwise margin loss from Section 3.2 to the downstream router logits $\mathbf{r}^{\mathrm{orig}}_{\ell+1}$ and $\mathbf{r}^{q}_{\ell+1}$. For each token t, let $\mathcal{P}_{\ell+1}(t)$ be the pair set constructed from the original downstream router ranking. Using the same definitions as above, we compute

$$z^{(t)}_{\ell+1,ij} = \eta m^{(t)}_{\ell+1,\mathrm{orig}}(i,j) - m^{(t)}_{\ell+1,q}(i,j). \tag{19}$$

The token-level downstream router loss is

$$\mathcal{L}^{(t)}_{\mathrm{dr}} = \frac{1}{|\mathcal{P}_{\ell+1}(t)|} \sum_{(i,j)\in\mathcal{P}_{\ell+1}(t)} \mathrm{softplus}(z^{(t)}_{\ell+1,ij}). \tag{20}$$

The mini-batch downstream loss is

$$\mathcal{L}_{\mathrm{dr}} = \frac{1}{|\mathcal{B}|} \sum_{t\in\mathcal{B}} \mathcal{L}^{(t)}_{\mathrm{dr}}. \tag{21}$$

Adaptive activation via top-k overlap. We apply the downstream loss only when block ℓ's quantization substantially changes the next router's top-k membership. At the start of block-ℓ training, we estimate a routing overlap score $\sigma_{\ell\to\ell+1}$ on a small probe set. For each token t, define

$$\mathcal{T}_{\mathrm{orig}}(t) = \mathrm{TopK}\Big(\mathbf{r}^{\mathrm{orig}}_{\ell+1}(t), k\Big), \tag{22}$$

and

$$\mathcal{T}_{q}(t) = \mathrm{TopK}\big(\mathbf{r}^{q}_{\ell+1}(t), k\big). \tag{23}$$

The block-level overlap is

$$\sigma_{\ell\to\ell+1} = \frac{1}{N_t k}\sum_t |\mathcal{T}_{\mathrm{orig}}(t)\cap\mathcal{T}_q(t)|, \quad \sigma_{\ell\to\ell+1}\in[0,1], \tag{24}$$

where N_t is the number of tokens in the probe set. We use a small probe set constructed by randomly sampling 32 examples from the calibration set. A value close to 1 means that the downstream top-k membership is mostly preserved, while a value close to 0 indicates severe routing mismatch.

We enable the downstream loss with the indicator

$$\mathbb{I}_{\mathrm{dr}}(\ell) = \mathbb{I}[\sigma_{\ell\to\ell+1} < \tau], \tag{25}$$

where τ is a model-specific tolerance threshold. If reconstruction already preserves the downstream top-k membership well, the downstream loss is skipped for that block. Empirically, we observed that applying the downstream loss to blocks whose next-router decisions are already well preserved does not always provide a useful optimization signal. In such cases, the remaining routing mismatch is small and the auxiliary loss may act more like an additional noisy constraint than a corrective signal. We therefore use the overlap threshold τ to apply the downstream loss only to blocks where quantization causes a sufficiently large change in downstream routing.

4. Experiments

4.1. Experimental Setup

Models. We evaluate DREAM-MoE on three open MoE LLMs: DeepSeek-V2-Lite-Chat (DeepSeek-AI, 2024), Moonlight-16B-A3B-Instruct (Liu et al., 2025), and Qwen3-30B-A3B (Yang et al., 2025). These models cover different MoE architectures and scales, allowing us to evaluate whether DREAM-MoE consistently improves routing stability across diverse MoE LLMs.

Quantization setup. We use AutoRound as the base learning-based PTQ framework. All experiments use weight-only quantization with group size 128 and symmetric quantization. We evaluate two low-bit settings. The first is standard 4-bit weight-only quantization. The second is a 3-bit-average setting following the EAC-MoE (Chen et al., 2025) setup: half of the experts closer to the model output are quantized to 2 bits so that the average weight precision over the full model becomes approximately 3 bits. Unless otherwise noted, all baselines use the same quantization configuration, calibration data, and optimization budget. For our method, the hyperparameters used in the experiments are summarized in Section A. Although DREAM-MoE is formulated as a backbone-independent auxiliary loss, this paper empirically validates its effectiveness only under the AutoRound instantiation; evaluation with other learning-based PTQ backbones is left for future work.

Table 1. Performance under 4-bit weight-only quantization. For each model, the best quantization method per benchmark is in **bold**.

Model	Method	PPL ↓	Accuracy ↑					
		WikiText-2	MMLU-Pro	GSM8K-CoT	GPQA-D	HumEval	IFEval	Avg.
DeepSeek-V2-Lite	Original	8.02	27.75	57.77	25.76	57.32	44.18	42.55
	AutoRound	8.37	25.24	56.41	23.74	51.83	40.11	39.46
	TopM-MSE	8.38	23.31	**58.07**	18.18	53.05	38.26	38.18
	ExpertQuant	8.74	19.39	42.68	22.22	42.68	35.49	32.49
	DREAM-MoE	**8.30**	**25.24**	55.80	**24.75**	**53.66**	**43.25**	**40.54**
Moonlight-16B-A3B	Original	8.46	37.01	76.72	28.79	73.17	44.73	52.09
	AutoRound	9.20	**35.52**	74.45	25.76	70.12	**42.33**	49.64
	TopM-MSE	9.16	33.44	75.13	22.73	69.51	41.04	48.37
	ExpertQuant	9.36	33.86	73.09	25.25	64.02	39.74	47.19
	DREAM-MoE	**9.13**	34.26	**75.36**	**30.30**	**70.12**	41.22	**50.25**
Qwen3-30B-A3B	Original	8.70	70.46	90.90	45.45	92.07	35.86	66.95
	AutoRound	**9.04**	69.23	89.69	35.35	91.46	34.57	64.06
	TopM-MSE	9.11	69.36	89.69	33.33	90.85	34.75	63.60
	ExpertQuant	9.40	67.26	88.93	37.88	88.41	35.12	63.52
	DREAM-MoE	9.08	**69.41**	**90.30**	**41.92**	**93.29**	**36.41**	**66.27**

Calibration data. For calibration, we use the Nemotron-Post-Training-Dataset-v1 released by NVIDIA (NVIDIA, 2025). We randomly sample 512 examples from the chat, code, math, and STEM subsets and use sequences of length 2,048. This calibration mixture provides reasoning-oriented data across natural language, programming, mathematics, and science.

Baselines. We compare DREAM-MoE against AutoRound and router-aware variants built on the same AutoRound backbone. The baselines include: (i) AutoRound, which uses only the standard block reconstruction objective; (ii) AutoRound with the TopM-MSE loss from EAC-MoE, where we set $M = 2k$ to include near-boundary experts beyond the routed top-k; and (iii) AutoRound with the router losses proposed by ExpertQuant, including Rank-Aware Jaccard (RAJ) Loss and Gap Hinge (GH) Loss.

Evaluation datasets. We evaluate DREAM-MoE on language modeling and downstream generation benchmarks. For language modeling, we report perplexity on WikiText-2 (Merity et al., 2017) following the GPTQ protocol (Frantar et al., 2023). For downstream evaluation, we use MMLU-Pro (Wang et al., 2024), GSM8K (Cobbe et al., 2021), GPQA-Diamond (Rein et al., 2024), IFEval (Zhou et al., 2023), and HumanEval (Chen et al., 2021). We run all downstream evaluations with `lm-evaluation-harness` (EleutherAI, 2026) using a vLLM backend (Kwon et al., 2023). Detailed evaluation metrics and generation-budget settings are provided in Section A.

4.2. Results

Language modeling. DREAM-MoE achieves the lowest WikiText-2 PPL in every (model, bit) configuration except Qwen3-30B-A3B at 4-bit, where AutoRound is ahead by only 0.04 PPL. Most strikingly, on Moonlight-16B-A3B at 3-bit our method reduces PPL from 15.46 (best baseline) to 13.93—a PPL reduction of more than 1.5 points that is disproportionately larger than the inter-method gap on any other configuration. Why our method yields such a dramatic benefit in this particular setting remains an open question.

Reasoning and instruction following. Averaged over MMLU-Pro, GSM8K-CoT, GPQA-D, HumanEval, and IFEval, DREAM-MoE ranks first in five of six (model, bit) configurations. The sole exception is DeepSeek-V2-Lite at 3-bit, where AutoRound leads by 1.76 points on average. The gain on Qwen3-30B-A3B is consistent at both bit-widths ($+2.21$ at 4-bit, $+1.08$ at 3-bit).

Code generation. On HumanEval, DREAM-MoE wins or ties in every model-bit configuration, including the 3-bit DeepSeek-V2-Lite case where it does not obtain the best average accuracy. This is consistent with code generation being highly brittle: small local errors in signatures, delimiters, indentation, or control-flow tokens can make a solution fail. Since quantization-induced expert shifts can alter the internal computation path over subsequent blocks and decoding steps, DREAM-MoE's downstream routing preservation may help maintain the local generation consistency required for code. We leave a detailed analysis of token-level losses and routing changes to future work.

5. Discussion

Margin preservation over TopM-MSE. As shown in Table 3, we isolate the effect of our margin-preserving router objective by removing the Downstream Routing Loss (DRL)

Table 2. Performance under 3-bit weight-only quantization. For each model, the best quantization method per benchmark is in **bold**.

Model	Method	PPL ↓	Accuracy ↑					
		WikiText-2	MMLU-Pro	GSM8K-CoT	GPQA-D	HumEval	IFEval	Avg.
DeepSeek-V2-Lite	Original	8.02	27.75	57.77	25.76	57.32	44.18	42.55
	AutoRound	10.76	**23.73**	48.98	**24.75**	37.20	**37.89**	**34.51**
	TopM-MSE	10.80	22.07	**50.27**	23.74	36.59	36.60	33.85
	ExpertQuant	11.33	15.96	36.24	20.20	30.49	33.46	27.27
	DREAM-MoE	**10.72**	21.24	45.19	21.72	**39.02**	36.60	32.75
Moonlight-16B-A3B	Original	8.46	37.01	76.72	28.79	73.17	44.73	52.09
	AutoRound	15.58	31.52	60.12	20.71	39.02	39.37	38.15
	TopM-MSE	15.46	**32.85**	59.67	**23.74**	42.68	39.19	39.63
	ExpertQuant	16.35	30.88	53.37	23.23	44.51	37.52	37.90
	DREAM-MoE	**13.93**	31.09	**60.73**	22.73	**48.78**	**40.30**	**40.72**
Qwen3-30B-A3B	Original	8.70	70.46	90.90	45.45	92.07	35.86	66.95
	AutoRound	11.48	**64.67**	89.08	31.31	84.15	33.09	60.46
	TopM-MSE	11.53	64.15	87.64	28.28	85.37	**34.94**	60.07
	ExpertQuant	12.05	61.47	86.96	29.80	83.54	34.57	59.27
	DREAM-MoE	**11.29**	63.80	**89.16**	**33.33**	**87.20**	34.20	**61.54**

Table 3. Ablation comparing TopM-MSE, DREAM-MoE w/o DRL, and DREAM-MoE w/ DRL. Avg denotes the mean accuracy across the five tasks.

Model	Bit	Method	PPL↓	Avg.↑
DeepSeek	4	TopM-MSE	8.38	38.18
		DREAM-MoE w/o DRL	8.31	39.97
		DREAM-MoE w/ DRL	**8.30**	**40.54**
	3	TopM-MSE	10.80	33.85
		DREAM-MoE w/o DRL	**10.72**	**34.27**
		DREAM-MoE w/ DRL	**10.72**	32.75
Moonlight	4	TopM-MSE	9.16	48.37
		DREAM-MoE w/o DRL	9.16	**51.52**
		DREAM-MoE w/ DRL	**9.13**	50.25
	3	TopM-MSE	15.46	39.63
		DREAM-MoE w/o DRL	15.84	38.89
		DREAM-MoE w/ DRL	**13.93**	**40.72**
Qwen3	4	TopM-MSE	9.11	63.60
		DREAM-MoE w/o DRL	**9.06**	65.67
		DREAM-MoE w/ DRL	9.08	**66.27**
	3	TopM-MSE	11.53	60.07
		DREAM-MoE w/o DRL	11.62	61.04
		DREAM-MoE w/ DRL	**11.29**	**61.54**

and comparing DREAM-MoE w/o DRL against TopM-MSE. TopM-MSE aligns router-logit values on the original top-M experts, but it does not directly preserve the pairwise margins that determine expert ordering and top-k membership. Replacing TopM-MSE with our margin-preserving loss improves average downstream accuracy in five of six settings, with an average gain of +1.68 points over the improved settings, even without DRL. PPL remains largely unchanged: it slightly improves at 4-bit but slightly degrades at 3-bit on Moonlight and Qwen3. This indicates that margin preservation mainly improves downstream task behavior, whereas DRL is needed to more consistently stabilize language-modeling fidelity.

Effect of DRL. Table 3 further shows the effect of adding DRL on top of the margin-preserving objective. Compared with DREAM-MoE w/o DRL, adding DRL improves average downstream accuracy in four of six configurations and reduces PPL in four configurations, with one additional tie on DeepSeek-V2-Lite at 3-bit. The largest PPL reduction appears on Moonlight-16B-A3B at 3-bit, where DRL reduces PPL from 15.84 to 13.93. These results indicate that DRL often provides an additional benefit by regularizing each quantized block with the downstream router behavior it induces. The effect is not uniform across all settings, as DRL slightly hurts average accuracy on Moonlight-16B-A3B at 4-bit and DeepSeek-V2-Lite at 3-bit, but overall it complements the margin-preserving objective and improves the final DREAM-MoE variant in most configurations.

6. Conclusion

We proposed DREAM-MoE, a margin-preserving auxiliary objective for MoE post-training quantization. DREAM-MoE preserves routing-critical expert orderings near the top-k boundary and further accounts for downstream routing errors induced by block-wise quantization. Across three MoE LLMs and two low-bit settings, DREAM-MoE improves average downstream accuracy and perplexity in most model-bit configurations without changing the inference-time model structure. These results indicate that preserving expert-selection margins and downstream routing behavior is a useful principle for accurate MoE quantization.

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Impact Statement

Mixture-of-Experts (MoE) architectures are increasingly adopted in frontier-scale foundation models, but most publicly available post-training quantization frameworks are still primarily designed for dense LLMs and do not fully address MoE-specific issues such as quantization-induced expert-shift. This work contributes toward closing that gap by improving the fidelity of low-bit MoE quantization.

This paper presents a post-training quantization method for MoE models. The proposed method does not introduce additional training data, new model capabilities, or new inference-time mechanisms. Its impact is primarily through reducing the memory and deployment cost of existing MoE models while better preserving their full-precision behavior. As such, we do not expect the method itself to raise additional ethical concerns beyond those of the underlying models being quantized.

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Table 4. Hyperparameters used for DREAM-MoE. DRL (%) denotes the percentage of MoE blocks for which the downstream-router loss is activated.

Model	Bits	λ_1	λ_2	τ	DRL (%)
DeepSeek-V2-Lite	4-bit	0.010	0.0010	0.865	15.4
	3-bit	0.010	0.0010	0.850	34.6
Moonlight-16B	4-bit	0.010	0.0010	0.925	11.5
	3-bit	0.010	0.0010	0.875	26.9
Qwen3-30B-A3B	4-bit	0.005	0.0005	0.955	14.9
	3-bit	0.005	0.0005	0.940	19.1

A. Additional Experimental Details

Hyperparameter selection. DREAM-MoE uses a small set of routing-loss hyperparameters. We keep the boundary-pool size fixed across all experiments with $b_{\mathrm{sel}} = 1$ and $b_{\mathrm{out}} = 2$, and use the same margin relaxation factor $\eta = 0.7$. The routing-loss weights λ_1 and λ_2 are chosen to keep the auxiliary routing losses smaller than the reconstruction loss while still affecting the learned quantization parameters.

The downstream routing threshold τ is selected per model and bit-width. We first define a candidate range based on the observed overlap between the original and quantized downstream top-k sets, and then select the value of τ that achieves the lowest validation perplexity. When the downstream router overlap varies widely, we use a coarser search interval of 0.025 to identify the active range. For models whose overlap distribution is concentrated near high values, such as Qwen3-30B-A3B, we use a finer interval of 0.005 to avoid over-activating the downstream loss. The selected thresholds and the resulting fractions of DRL-activated MoE blocks are reported in Table 4.

Evaluation details and reasoning budgets. For WikiText-2 perplexity (Merity et al., 2017), we use non-overlapping 2,048-token chunks following the GPTQ protocol (Frantar et al., 2023). For downstream tasks, we use `lm-evaluation-harness` (EleutherAI, 2026) with a vLLM backend (Kwon et al., 2023) and apply the chat template for all instruction-tuned models. All reasoning and instruction-following benchmarks are run with greedy decoding, i.e., temperature 0. MMLU-Pro is evaluated in a 5-shot setting, while GSM8K-CoT, GPQA-Diamond, IFEval, and HumanEval are evaluated zero-shot. We report `exact match` for MMLU-Pro, GSM8K, and GPQA-Diamond, `prompt level strict acc` for IFEval, and `pass@1` for HumanEval.

The maximum number of generated tokens is $3{,}072$ for every downstream benchmark except GPQA-Diamond, where we allow up to $8{,}192$ tokens to accommodate longer chain-of-thought responses for graduate-level science questions. Because Moonlight-16B-A3B has an $8{,}192$-token context window, its GPQA-Diamond generation budget is capped at $4{,}096$ tokens so that the prompt and response together fit within the context limit; DeepSeek-V2-Lite-Chat and Qwen3-30B-A3B receive the full $8{,}192$-token budget. All other tasks fit within Moonlight's context window under the $3{,}072$-token budget.